

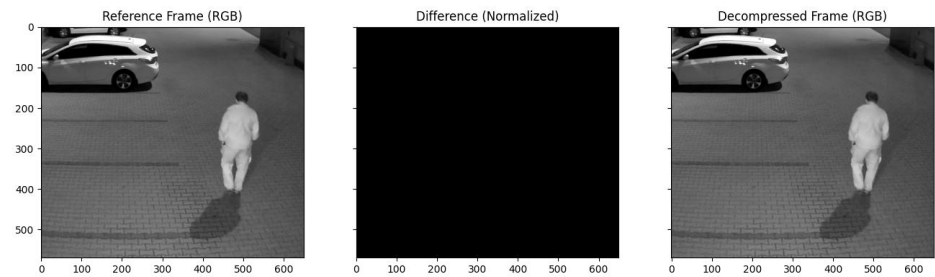
Sprawozdanie z laboratorium nr.9

Sebastian Marcinowski

Compression Results

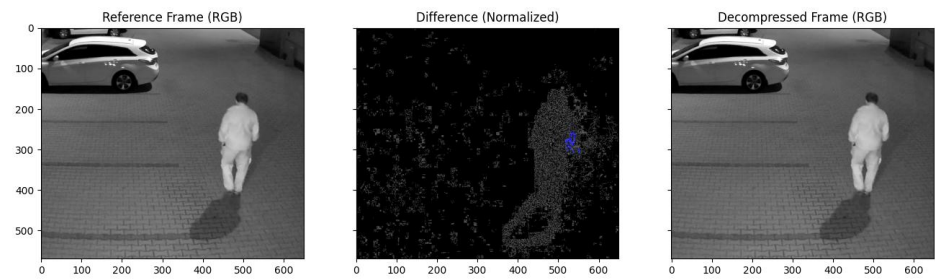
Subsampling: 4:4:4, Divider: 1, Method: None

File:clip_3.mp4, subsampling=4:4:4, divider=1, KeyFrame=4



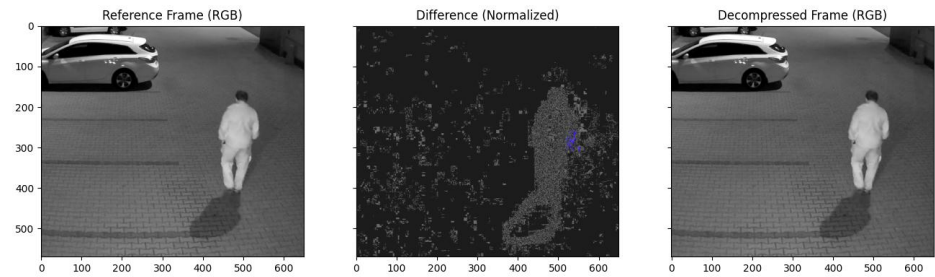
Subsampling: 4:4:4, Divider: 2, Method: None

File:clip_3.mp4, subsampling=4:4:4, divider=2, KeyFrame=4



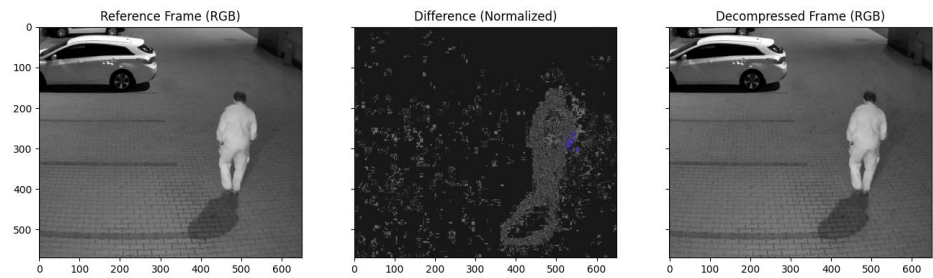
Subsampling: 4:4:4, Divider: 4, Method: None

File:clip_3.mp4, subsampling=4:4:4, divider=4, KeyFrame=4



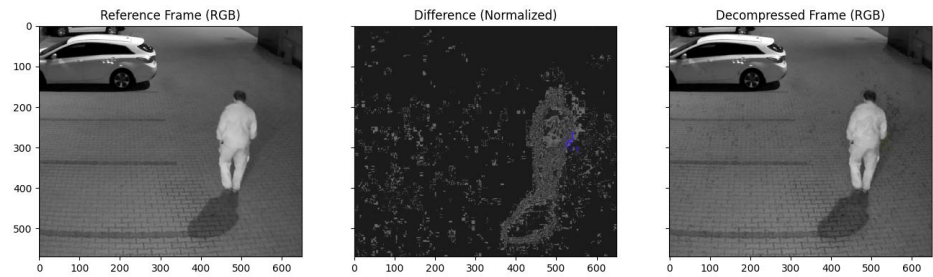
Subsampling: 4:4:4, Divider: 8, Method: None

File:clip_3.mp4, subsampling=4:4:4, divider=8, KeyFrame=4



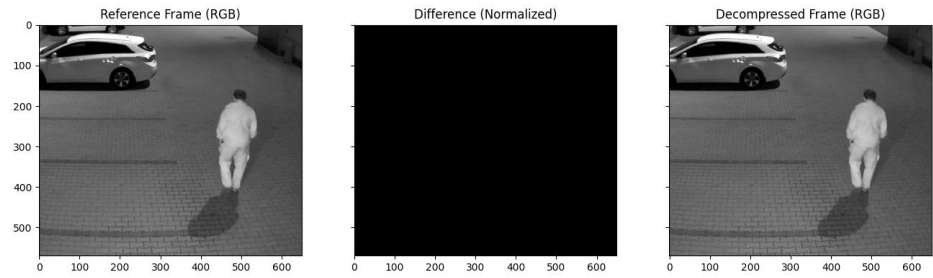
Subsampling: 4:4:4, Divider: 16, Method: None

File:clip_3.mp4, subsampling=4:4:4, divider=16, KeyFrame=4



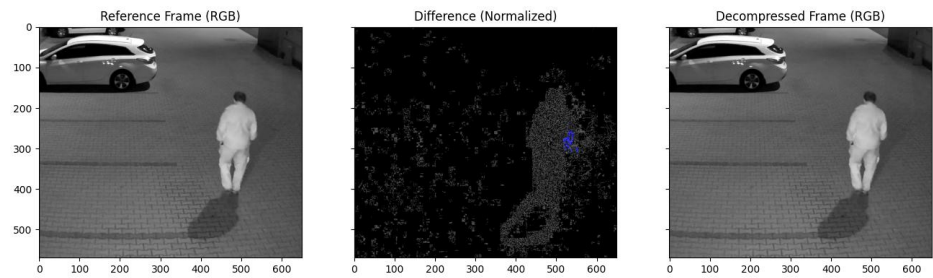
Subsampling: 4:2:2, Divider: 1, Method: None

File:clip_3.mp4, subsampling=4:2:2, divider=1, KeyFrame=4



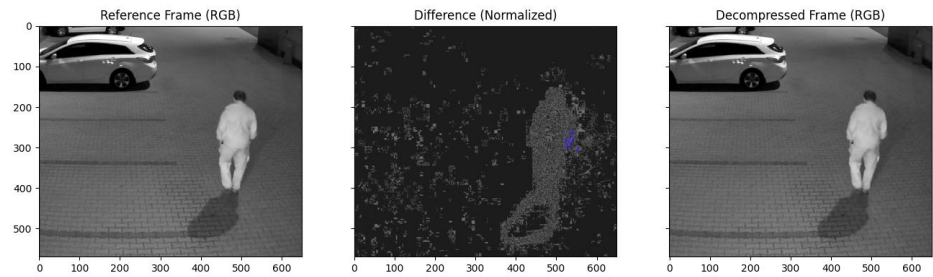
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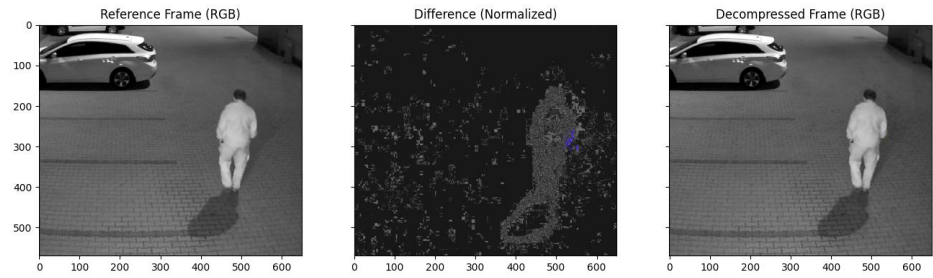
Subsampling: 4:2:2, Divider: 4, Method: None

File:clip_3.mp4, subsampling=4:2:2, divider=4, KeyFrame=4



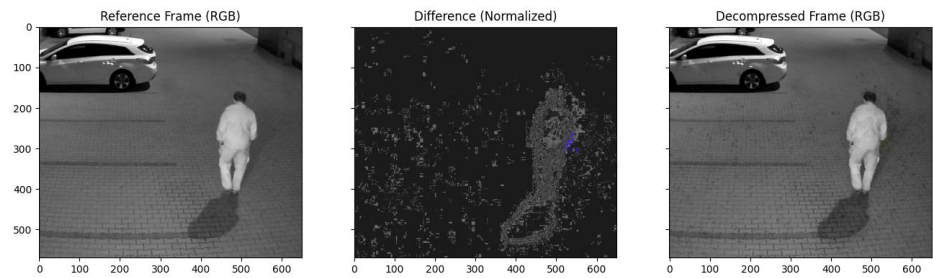
Subsampling: 4:2:2, Divider: 8, Method: None

File:clip_3.mp4, subsampling=4:2:2, divider=8, KeyFrame=4



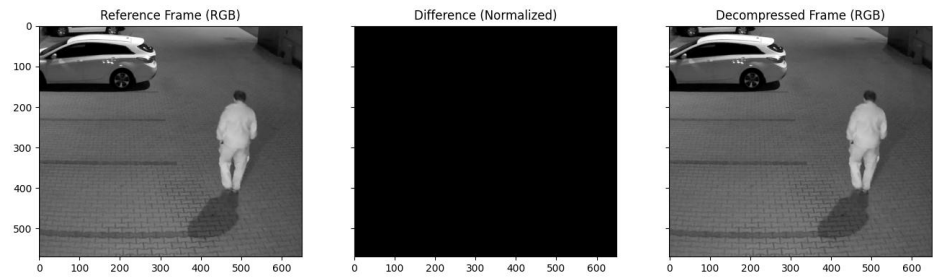
Subsampling: 4:2:2, Divider: 16, Method: None

File:clip_3.mp4, subsampling=4:2:2, divider=16, KeyFrame=4



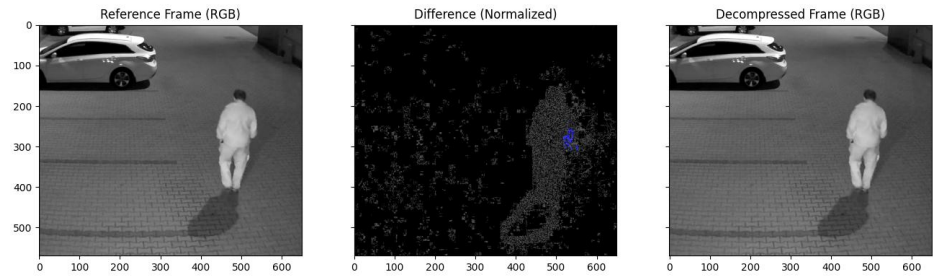
Subsampling: 4:4:0, Divider: 1, Method: None

File:clip_3.mp4, subsampling=4:4:0, divider=1, KeyFrame=4



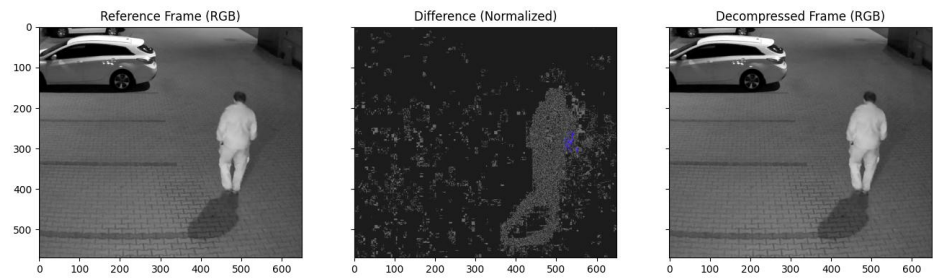
Subsampling: 4:4:0, Divider: 2, Method: None

File:clip_3.mp4, subsampling=4:4:0, divider=2, KeyFrame=4



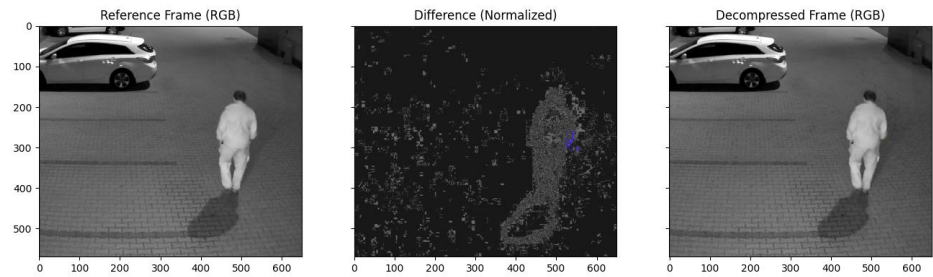
Subsampling: 4:4:0, Divider: 4, Method: None

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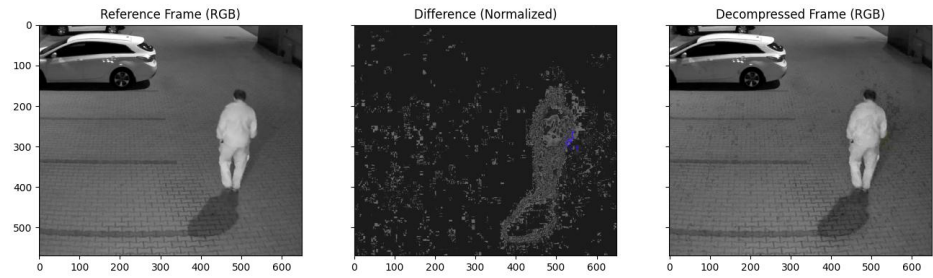
Subsampling: 4:4:0, Divider: 8, Method: None

File:clip_3.mp4, subsampling=4:4:0, divider=8, KeyFrame=4



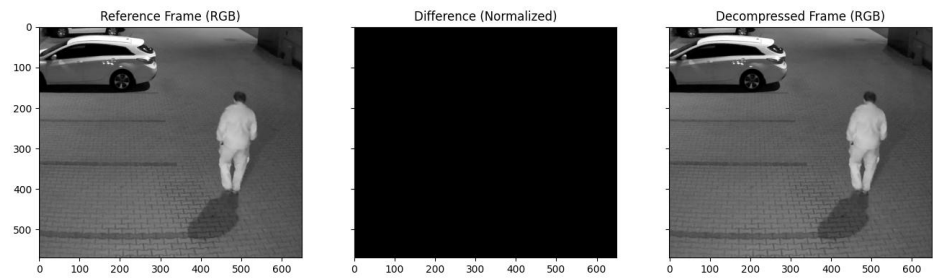
Subsampling: 4:4:0, Divider: 16, Method: None

File:clip_3.mp4, subsampling=4:4:0, divider=16, KeyFrame=4



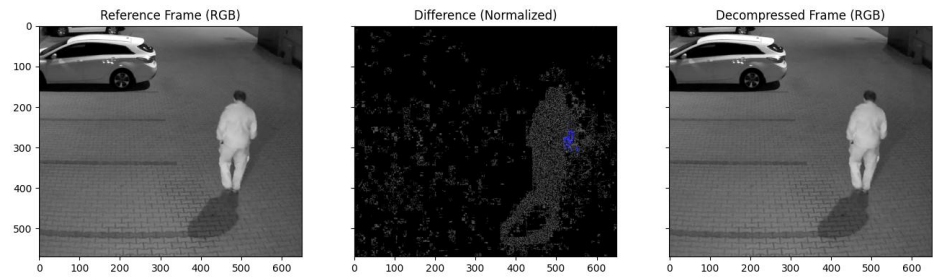
Subsampling: 4:2:0, Divider: 1, Method: None

File:clip_3.mp4, subsampling=4:2:0, divider=1, KeyFrame=4



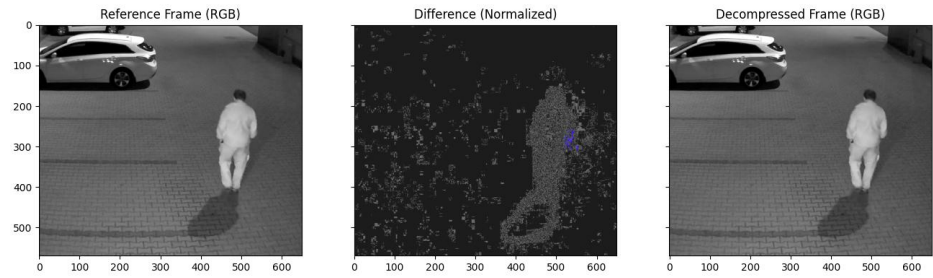
Subsampling: 4:2:0, Divider: 2, Method: None

File:clip_3.mp4, subsampling=4:2:0, divider=2, KeyFrame=4



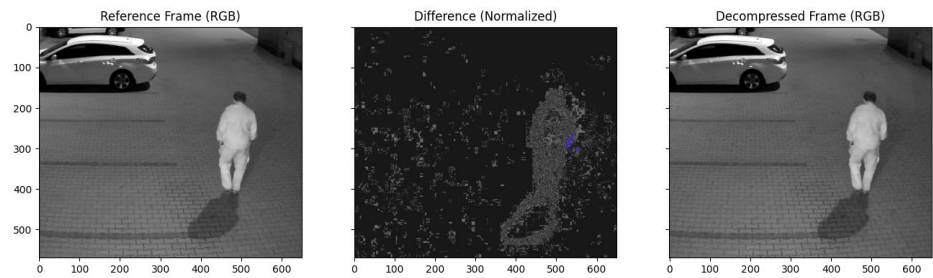
Subsampling: 4:2:0, Divider: 4, Method: None

File:clip_3.mp4, subsampling=4:2:0, divider=4, KeyFrame=4



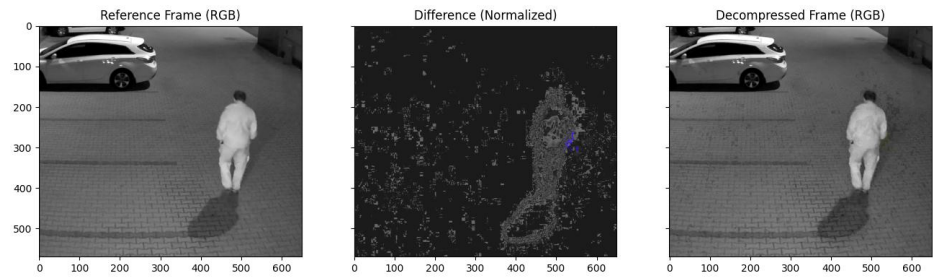
Subsampling: 4:2:0, Divider: 8, Method: None

File:clip_3.mp4, subsampling=4:2:0, divider=8, KeyFrame=4



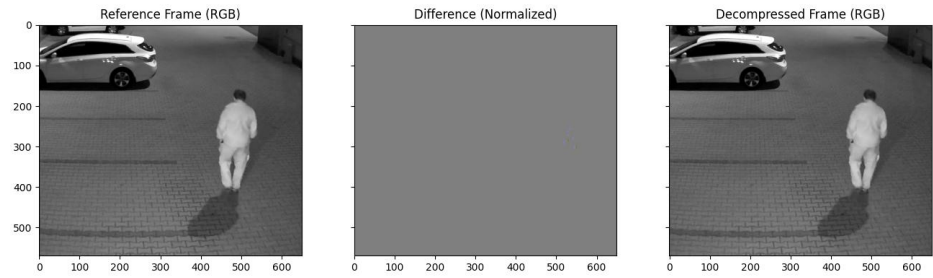
Subsampling: 4:2:0, Divider: 16, Method: None

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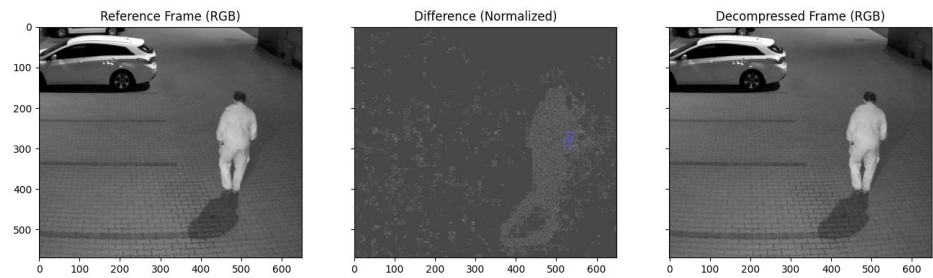
Subsampling: 4:1:1, Divider: 1, Method: None

File:clip_3.mp4, subsampling=4:1:1, divider=1, KeyFrame=4



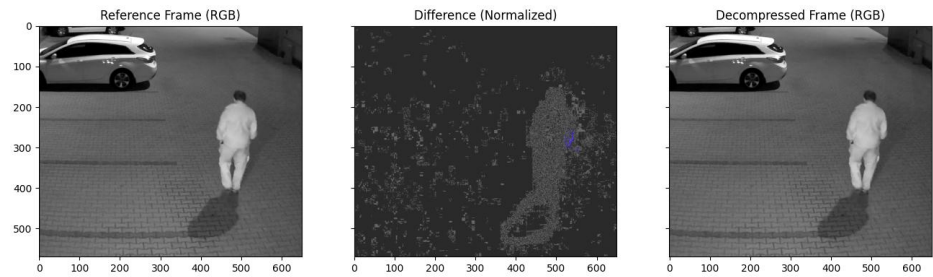
Subsampling: 4:1:1, Divider: 2, Method: None

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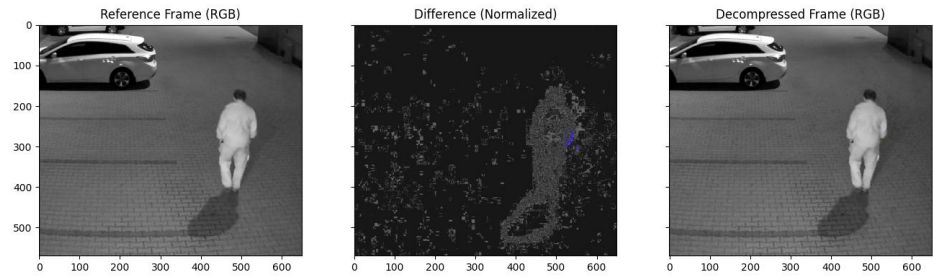
Subsampling: 4:1:1, Divider: 4, Method: None

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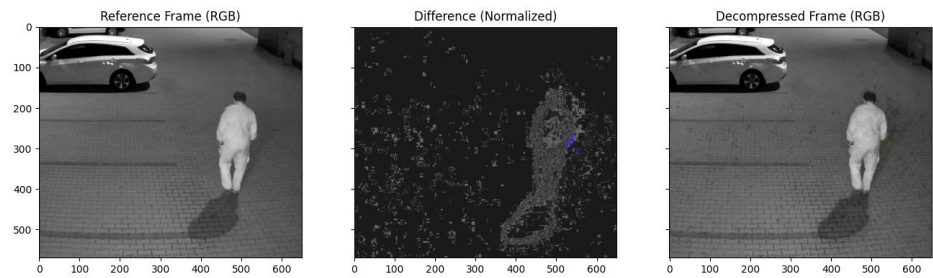
Subsampling: 4:1:1, Divider: 8, Method: None

File:clip_3.mp4, subsampling=4:1:1, divider=8, KeyFrame=4



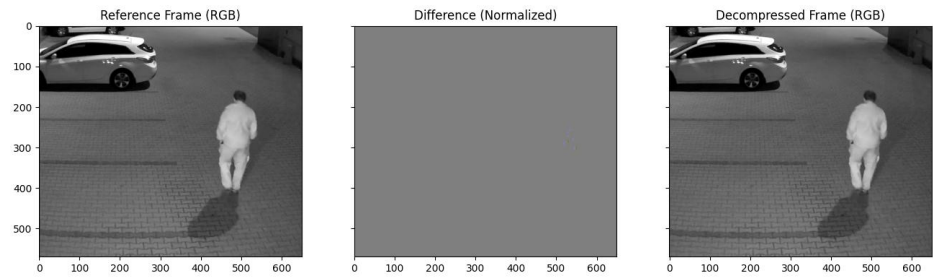
Subsampling: 4:1:1, Divider: 16, Method: None

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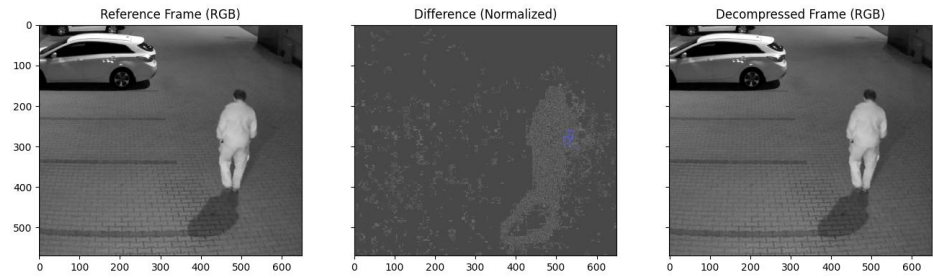
Subsampling: 4:1:0, Divider: 1, Method: None

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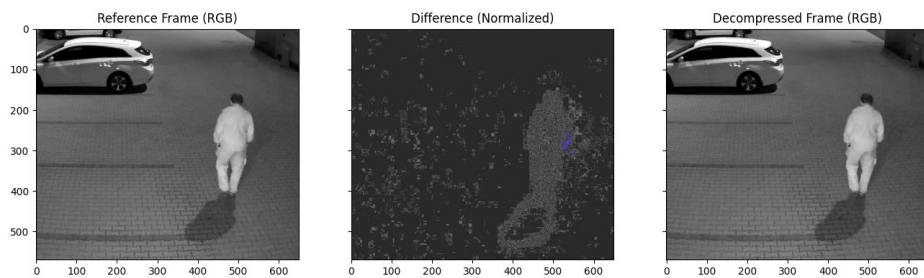
Subsampling: 4:1:0, Divider: 2, Method: None

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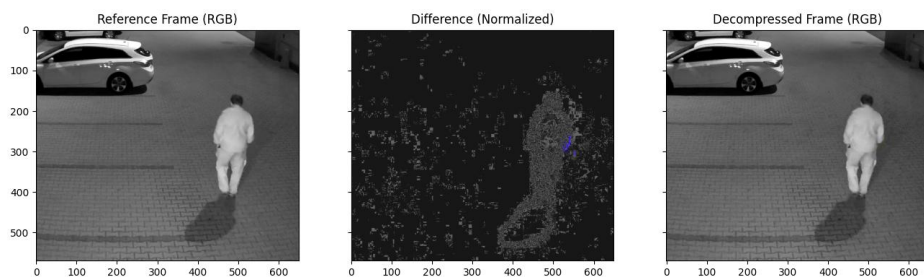
Subsampling: 4:1:0, Divider: 4, Method: None

File:clip_3.mp4, subsampling=4:1:0, divider=4, KeyFrame=4



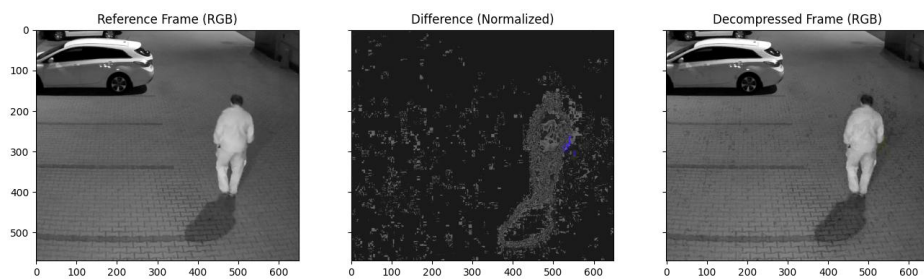
Subsampling: 4:1:0, Divider: 8, Method: None

File:clip_3.mp4, subsampling=4:1:0, divider=8, KeyFrame=4



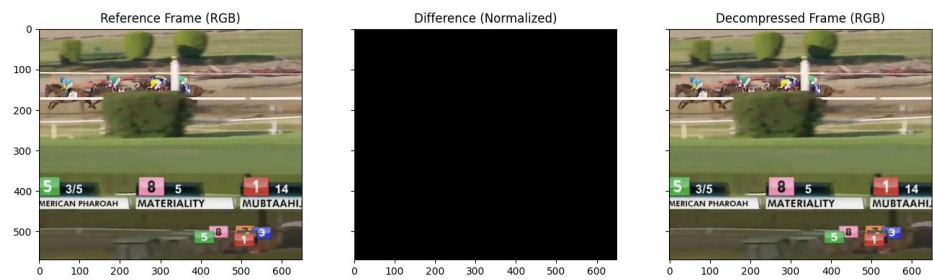
Subsampling: 4:1:0, Divider: 16, Method: None

File:clip_3.mp4, subsampling=4:1:0, divider=16, KeyFrame=4



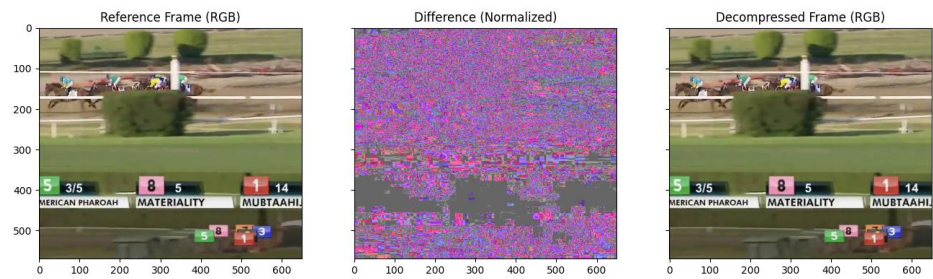
Subsampling: 4:4:4, Divider: 1, Method: None

File:clip_4.mp4, subsampling=4:4:4, divider=1, KeyFrame=4



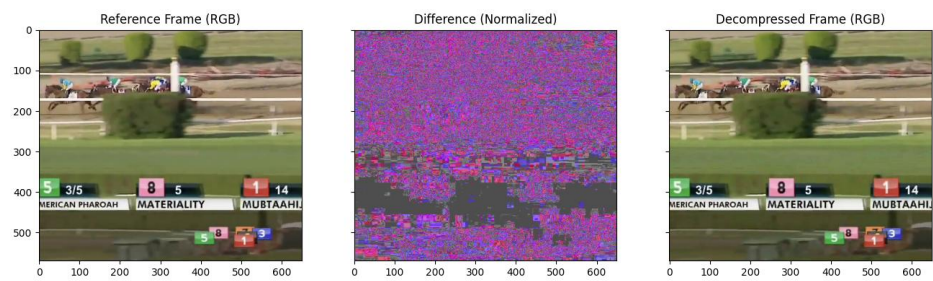
Subsampling: 4:4:4, Divider: 2, Method: None

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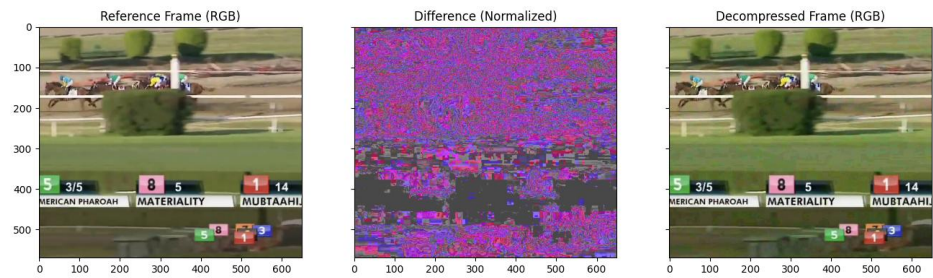
Subsampling: 4:4:4, Divider: 4, Method: None

File:clip_4.mp4, subsampling=4:4:4, divider=4, KeyFrame=4



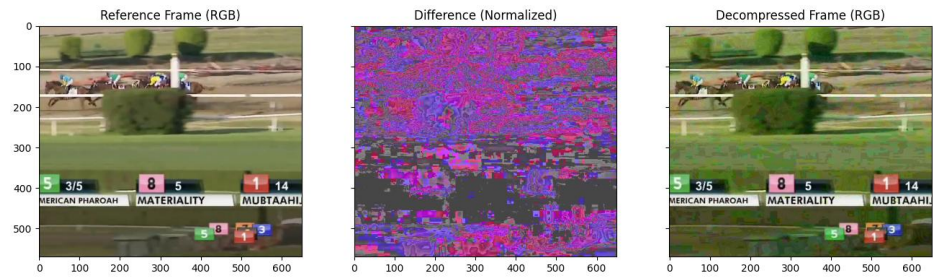
Subsampling: 4:4:4, Divider: 8, Method: None

File:clip_4.mp4, subsampling=4:4:4, divider=8, KeyFrame=4



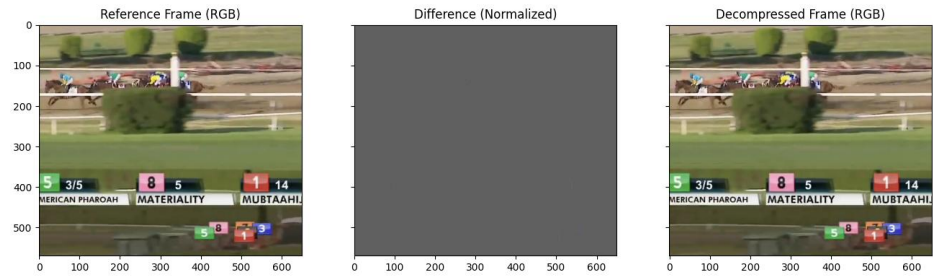
Subsampling: 4:4:4, Divider: 16, Method: None

File:clip_4.mp4, subsampling=4:4:4, divider=16, KeyFrame=4



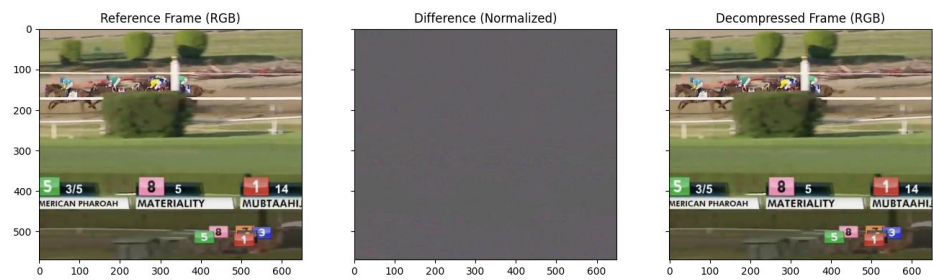
Subsampling: 4:2:2, Divider: 1, Method: None

File:clip_4.mp4, subsampling=4:2:2, divider=1, KeyFrame=4



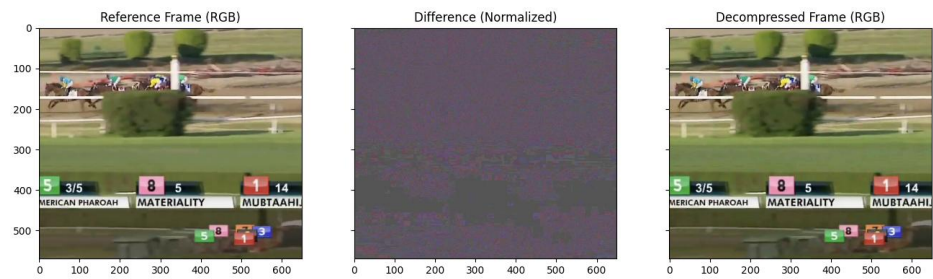
Subsampling: 4:2:2, Divider: 2, Method: None

File:clip_4.mp4, subsampling=4:2:2, divider=2, KeyFrame=4



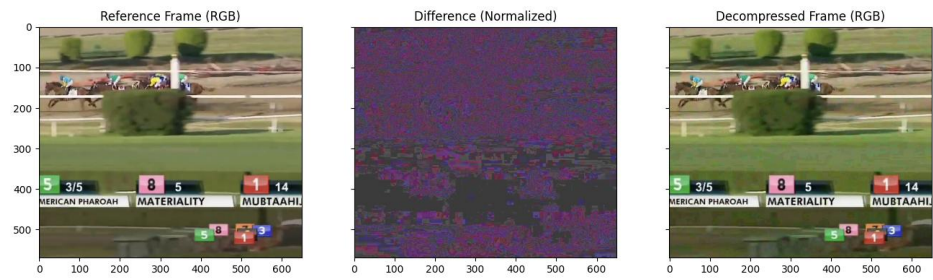
Subsampling: 4:2:2, Divider: 4, Method: None

File:clip_4.mp4, subsampling=4:2:2, divider=4, KeyFrame=4



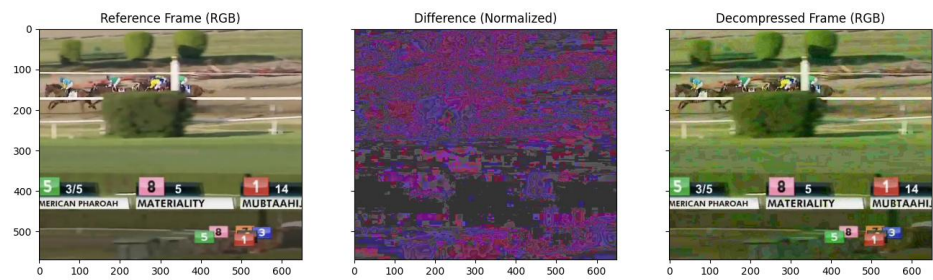
Subsampling: 4:2:2, Divider: 8, Method: None

File:clip_4.mp4, subsampling=4:2:2, divider=8, KeyFrame=4



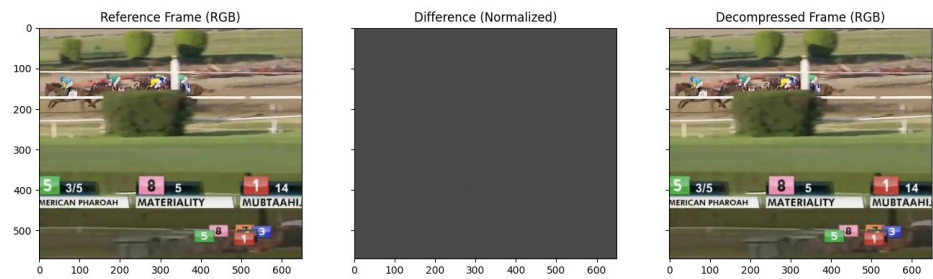
Subsampling: 4:2:2, Divider: 16, Method: None

File:clip_4.mp4, subsampling=4:2:2, divider=16, KeyFrame=4



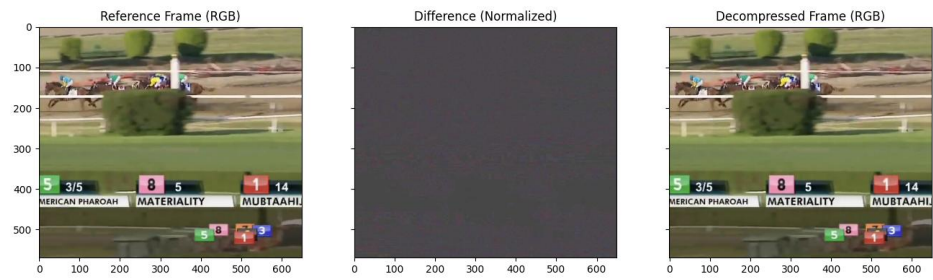
Subsampling: 4:4:0, Divider: 1, Method: None

File:clip_4.mp4, subsampling=4:4:0, divider=1, KeyFrame=4



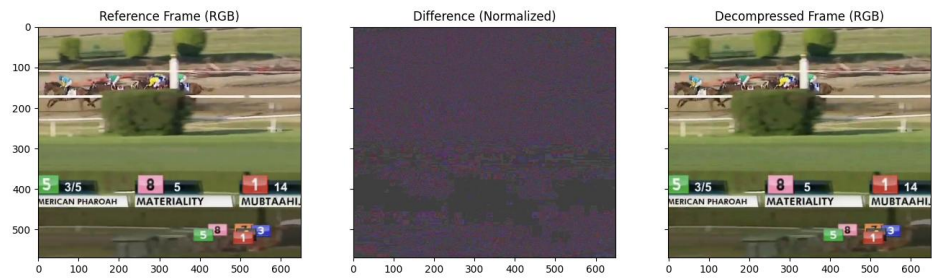
Subsampling: 4:4:0, Divider: 2, Method: None

File:clip_4.mp4, subsampling=4:4:0, divider=2, KeyFrame=4



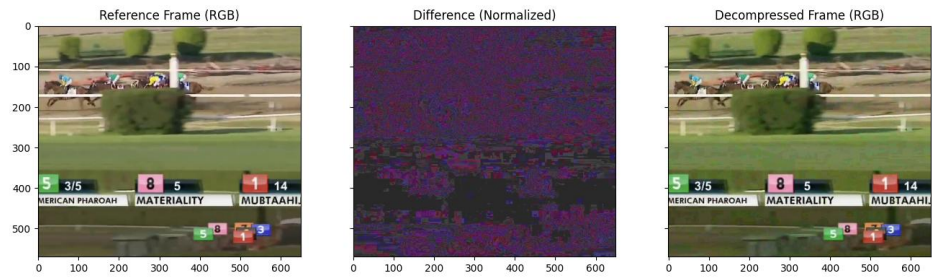
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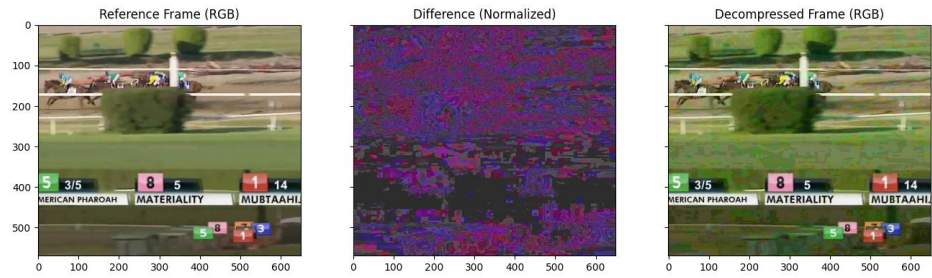
Subsampling: 4:4:0, Divider: 8, Method: None

File:clip_4.mp4, subsampling=4:4:0, divider=8, KeyFrame=4



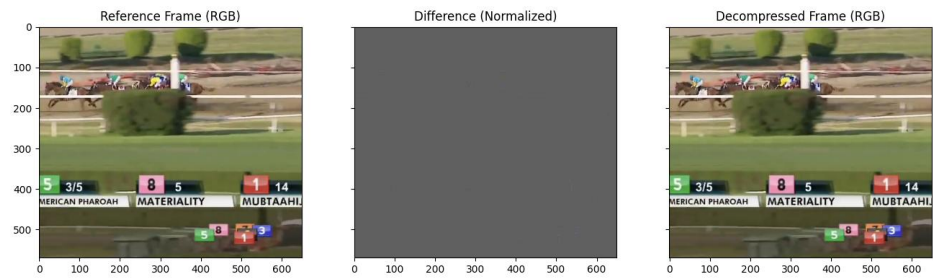
Subsampling: 4:4:0, Divider: 16, Method: None

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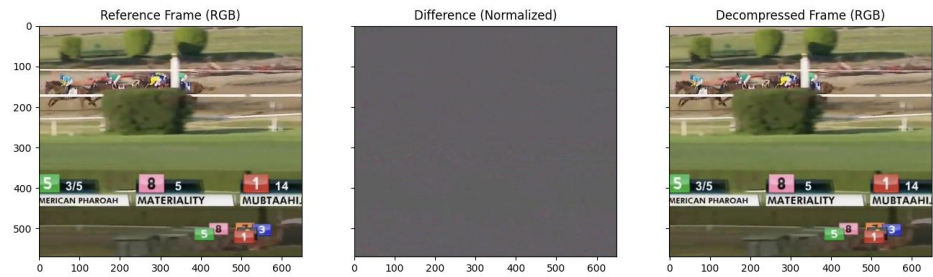
Subsampling: 4:2:0, Divider: 1, Method: None

File:clip_4.mp4, subsampling=4:2:0, divider=1, KeyFrame=4



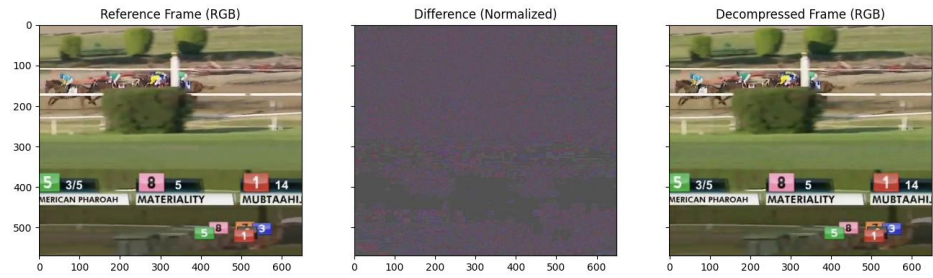
Subsampling: 4:2:0, Divider: 2, Method: None

File:clip_4.mp4, subsampling=4:2:0, divider=2, KeyFrame=4



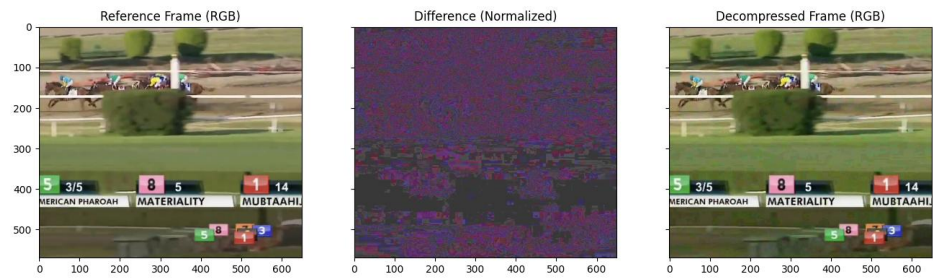
Subsampling: 4:2:0, Divider: 4, Method: None

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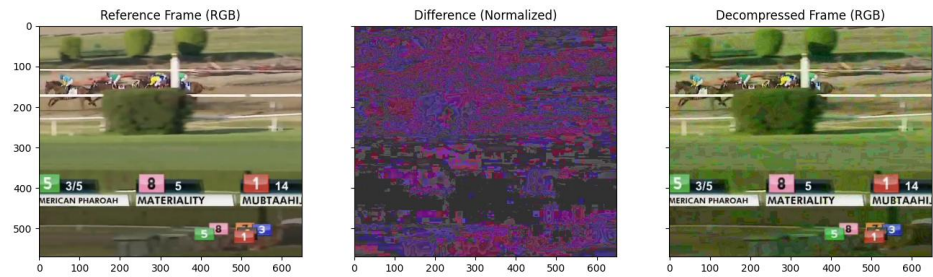
Subsampling: 4:2:0, Divider: 8, Method: None

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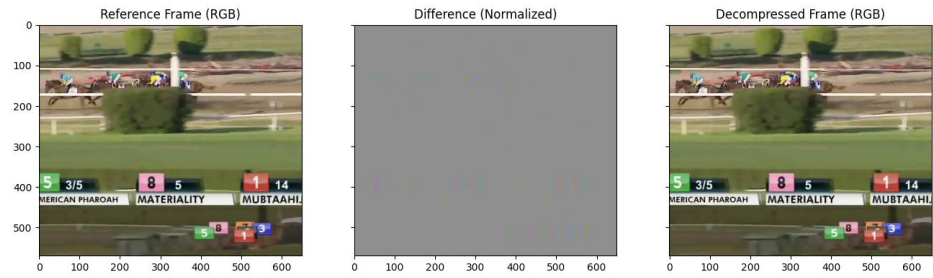
Subsampling: 4:2:0, Divider: 16, Method: None

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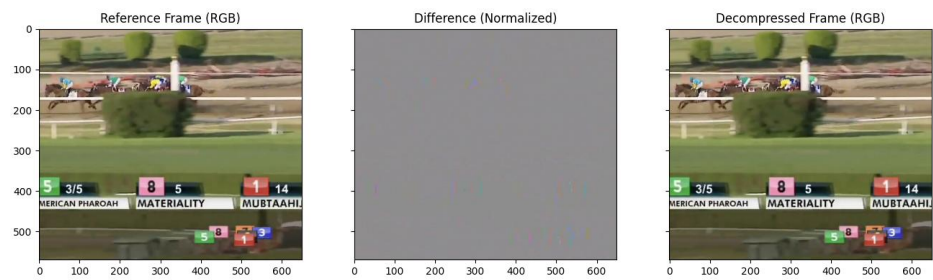
Subsampling: 4:1:1, Divider: 1, Method: None

File:clip_4.mp4, subsampling=4:1:1, divider=1, KeyFrame=4



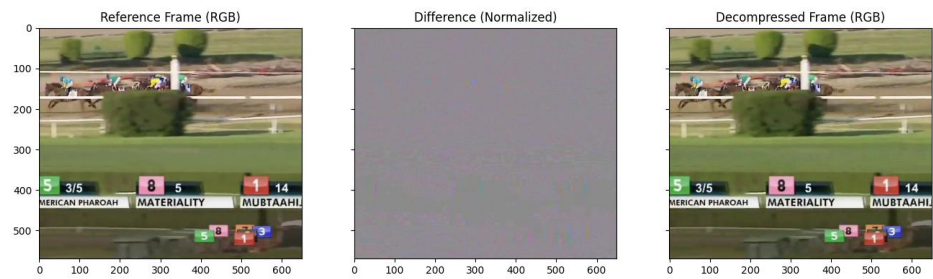
Subsampling: 4:1:1, Divider: 2, Method: None

File:clip_4.mp4, subsampling=4:1:1, divider=2, KeyFrame=4



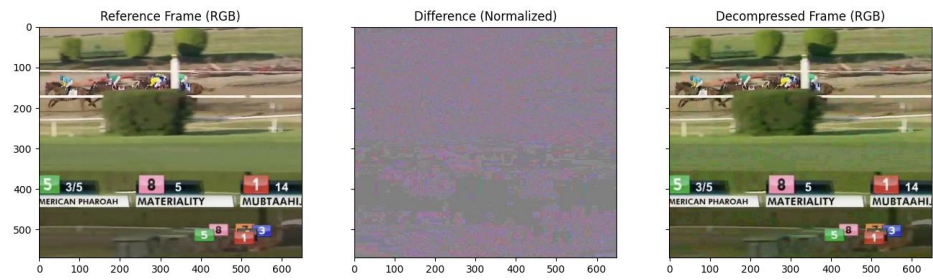
Subsampling: 4:1:1, Divider: 4, Method: None

File:clip_4.mp4, subsampling=4:1:1, divider=4, KeyFrame=4



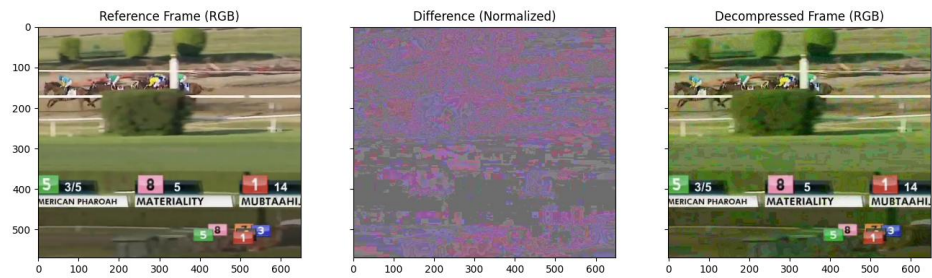
Subsampling: 4:1:1, Divider: 8, Method: None

File:clip_4.mp4, subsampling=4:1:1, divider=8, KeyFrame=4



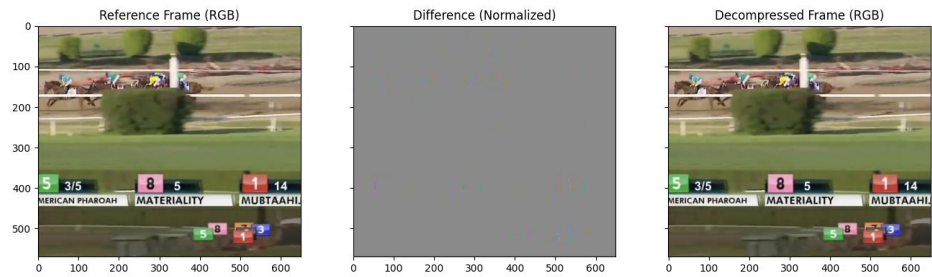
Subsampling: 4:1:1, Divider: 16, Method: None

File:clip_4.mp4, subsampling=4:1:1, divider=16, KeyFrame=4



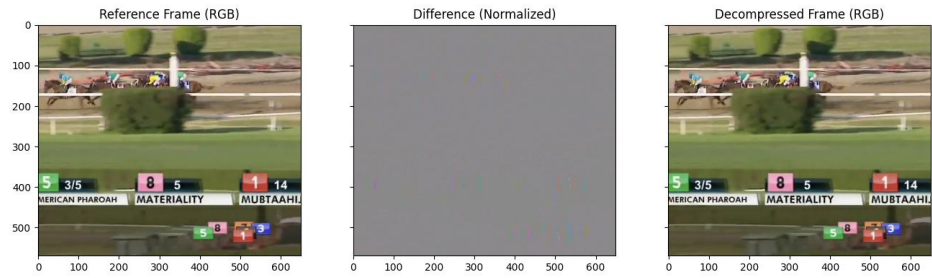
Subsampling: 4:1:0, Divider: 1, Method: None

File:clip_4.mp4, subsampling=4:1:0, divider=1, KeyFrame=4



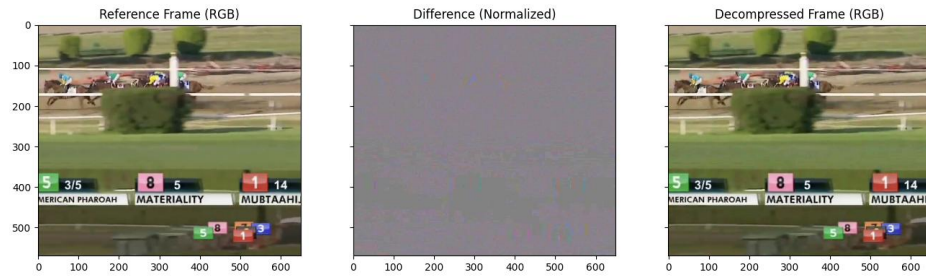
Subsampling: 4:1:0, Divider: 2, Method: None

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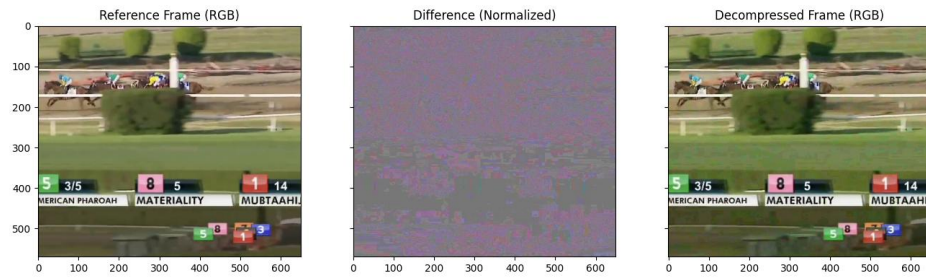
Subsampling: 4:1:0, Divider: 4, Method: None

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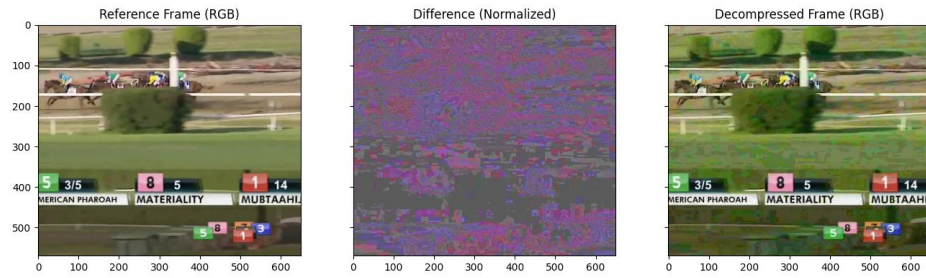
Subsampling: 4:1:0, Divider: 8, Method: None

File:clip_4.mp4, subsampling=4:1:0, divider=8, KeyFrame=4



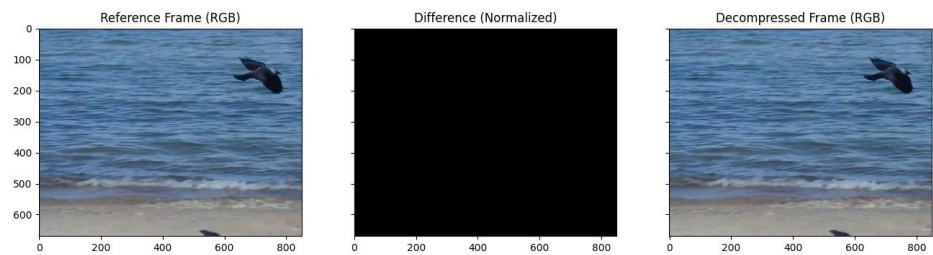
Subsampling: 4:1:0, Divider: 16, Method: None

File:clip_4.mp4, subsampling=4:1:0, divider=16, KeyFrame=4



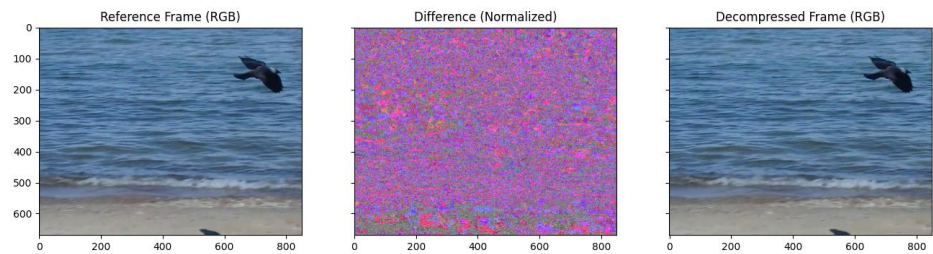
Subsampling: 4:4:4, Divider: 1, Method: None

File:clip_5.mp4, subsampling=4:4:4, divider=1, KeyFrame=4



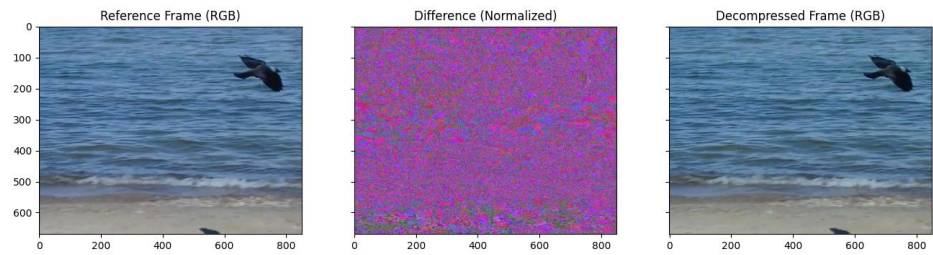
Subsampling: 4:4:4, Divider: 2, Method: None

File:clip_5.mp4, subsampling=4:4:4, divider=2, KeyFrame=4



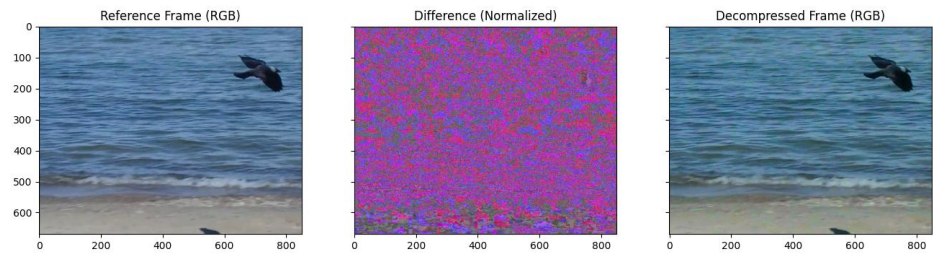
Subsampling: 4:4:4, Divider: 4, Method: None

File:clip_5.mp4, subsampling=4:4:4, divider=4, KeyFrame=4



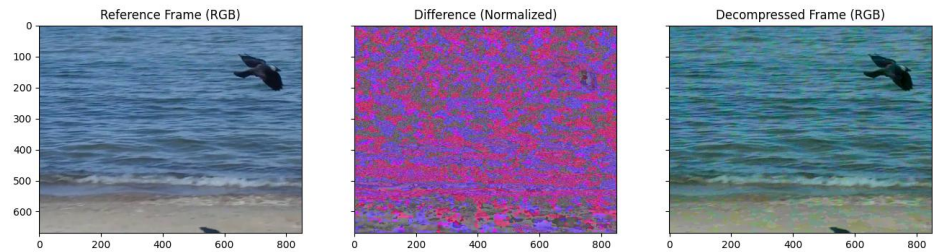
Subsampling: 4:4:4, Divider: 8, Method: None

File:clip_5.mp4, subsampling=4:4:4, divider=8, KeyFrame=4



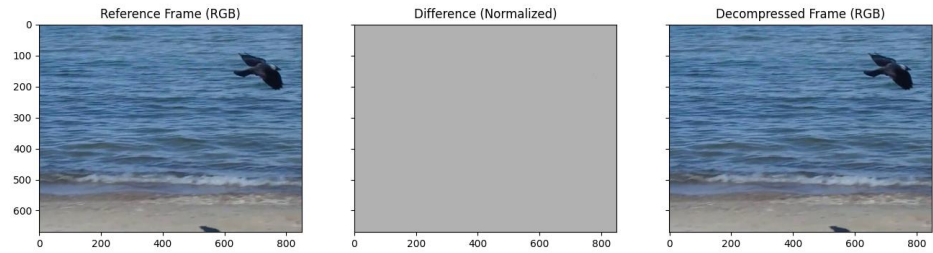
Subsampling: 4:4:4, Divider: 16, Method: None

File:clip_5.mp4, subsampling=4:4:4, divider=16, KeyFrame=4



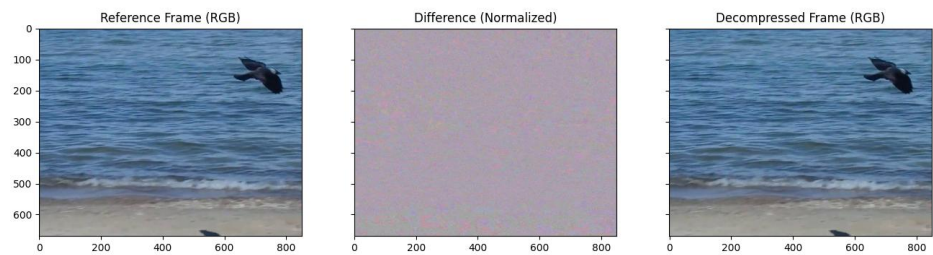
Subsampling: 4:2:2, Divider: 1, Method: None

File:clip_5.mp4, subsampling=4:2:2, divider=1, KeyFrame=4



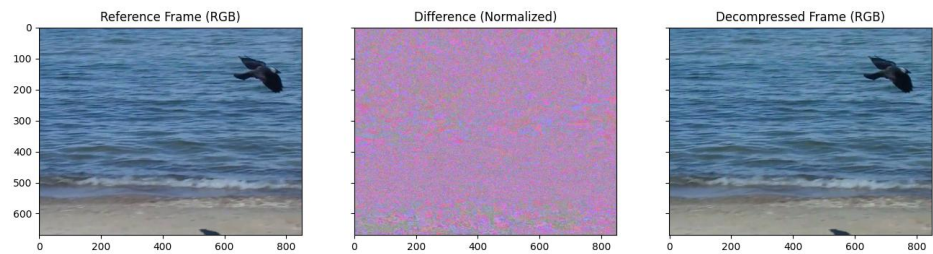
Subsampling: 4:2:2, Divider: 2, Method: None

File:clip_5.mp4, subsampling=4:2:2, divider=2, KeyFrame=4



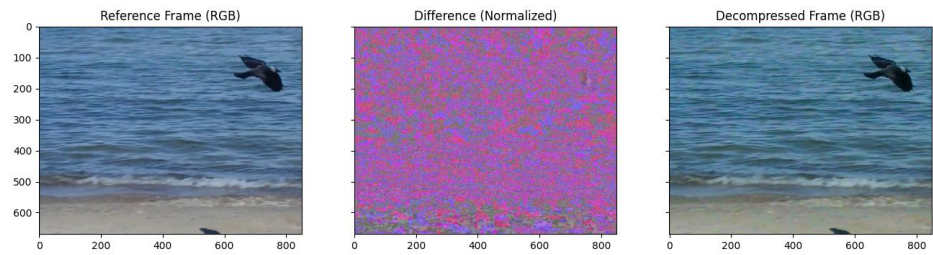
Subsampling: 4:2:2, Divider: 4, Method: None

File:clip_5.mp4, subsampling=4:2:2, divider=4, KeyFrame=4



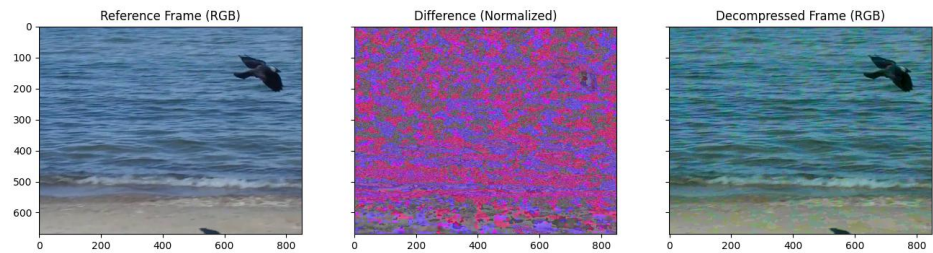
Subsampling: 4:2:2, Divider: 8, Method: None

File:clip_5.mp4, subsampling=4:2:2, divider=8, KeyFrame=4



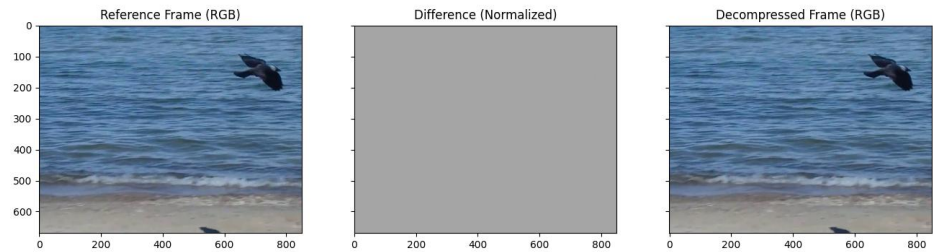
Subsampling: 4:2:2, Divider: 16, Method: None

File:clip_5.mp4, subsampling=4:2:2, divider=16, KeyFrame=4



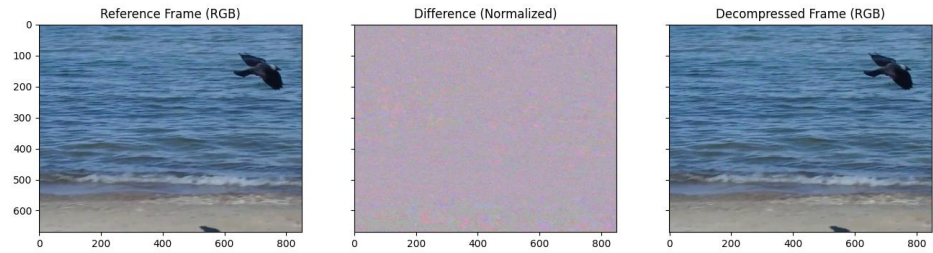
Subsampling: 4:4:0, Divider: 1, Method: None

File:clip_5.mp4, subsampling=4:4:0, divider=1, KeyFrame=4



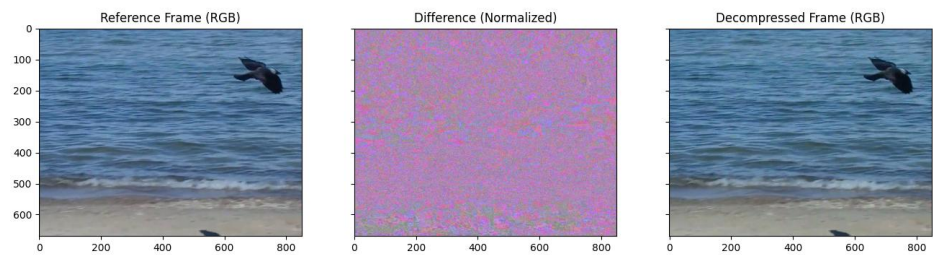
Subsampling: 4:4:0, Divider: 2, Method: None

File:clip_5.mp4, subsampling=4:4:0, divider=2, KeyFrame=4



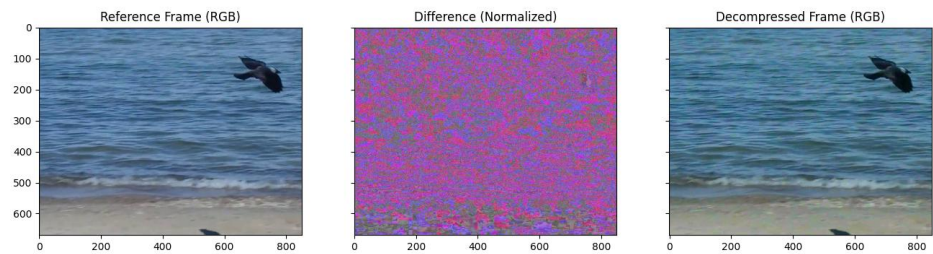
Subsampling: 4:4:0, Divider: 4, Method: None

File:clip_5.mp4, subsampling=4:4:0, divider=4, KeyFrame=4



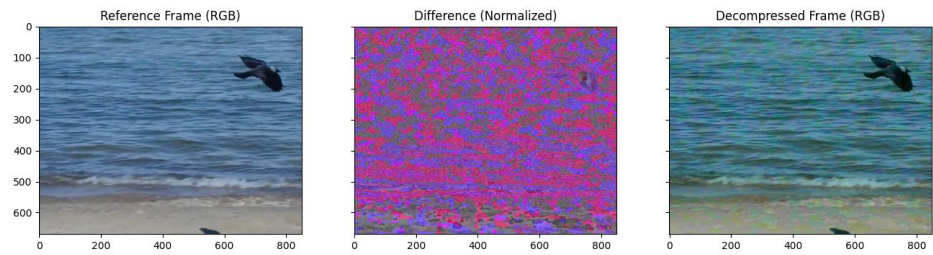
Subsampling: 4:4:0, Divider: 8, Method: None

File:clip_5.mp4, subsampling=4:4:0, divider=8, KeyFrame=4



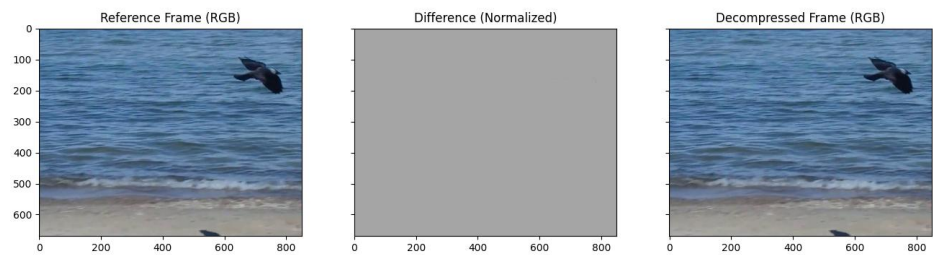
Subsampling: 4:4:0, Divider: 16, Method: None

File:clip_5.mp4, subsampling=4:4:0, divider=16, KeyFrame=4



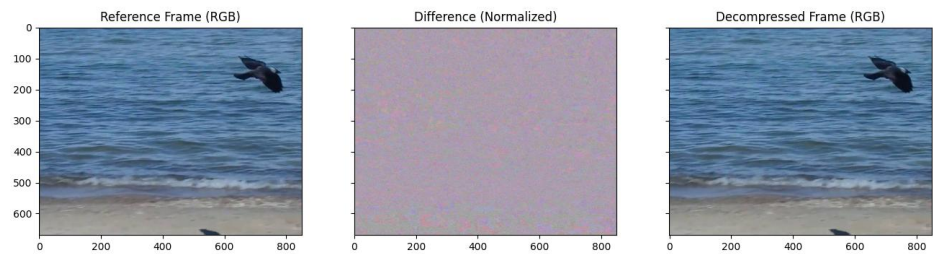
Subsampling: 4:2:0, Divider: 1, Method: None

File:clip_5.mp4, subsampling=4:2:0, divider=1, KeyFrame=4



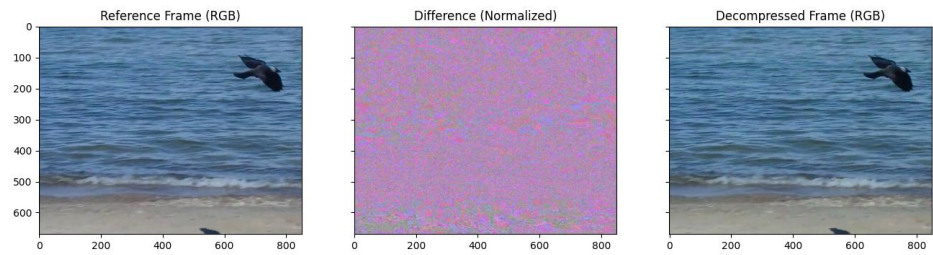
Subsampling: 4:2:0, Divider: 2, Method: None

File:clip_5.mp4, subsampling=4:2:0, divider=2, KeyFrame=4



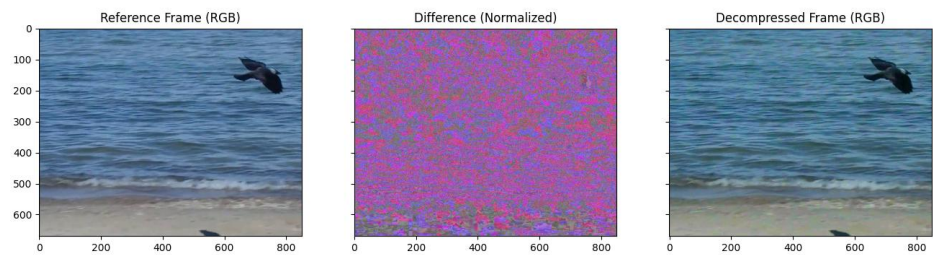
Subsampling: 4:2:0, Divider: 4, Method: None

File:clip_5.mp4, subsampling=4:2:0, divider=4, KeyFrame=4



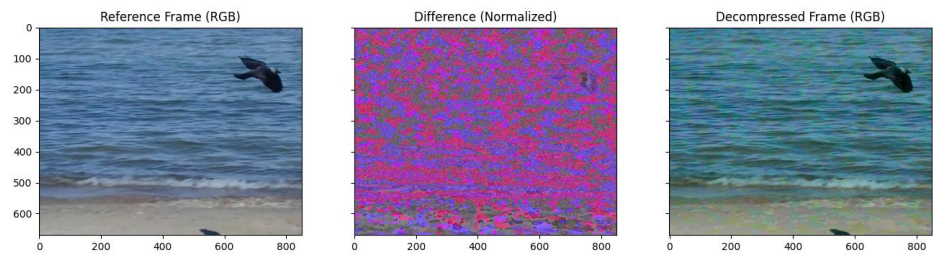
Subsampling: 4:2:0, Divider: 8, Method: None

File:clip_5.mp4, subsampling=4:2:0, divider=8, KeyFrame=4



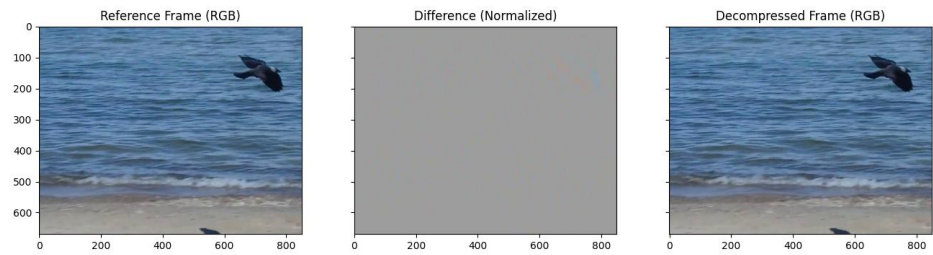
Subsampling: 4:2:0, Divider: 16, Method: None

File:clip_5.mp4, subsampling=4:2:0, divider=16, KeyFrame=4



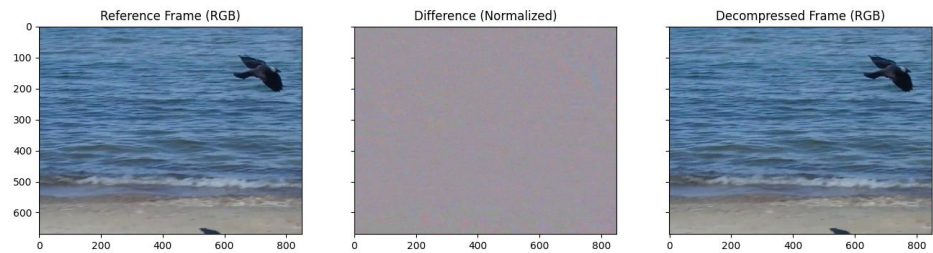
Subsampling: 4:1:1, Divider: 1, Method: None

File:clip_5.mp4, subsampling=4:1:1, divider=1, KeyFrame=4



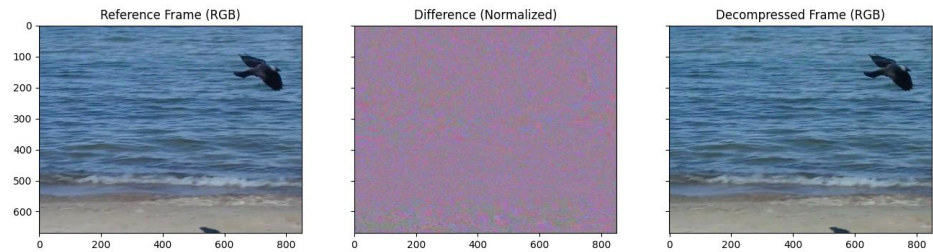
Subsampling: 4:1:1, Divider: 2, Method: None

File:clip_5.mp4, subsampling=4:1:1, divider=2, KeyFrame=4



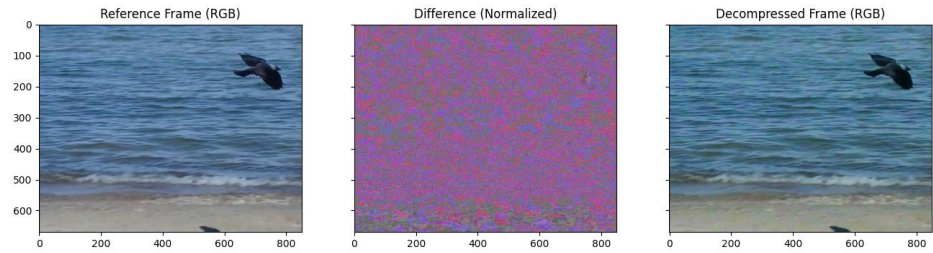
Subsampling: 4:1:1, Divider: 4, Method: None

File:clip_5.mp4, subsampling=4:1:1, divider=4, KeyFrame=4



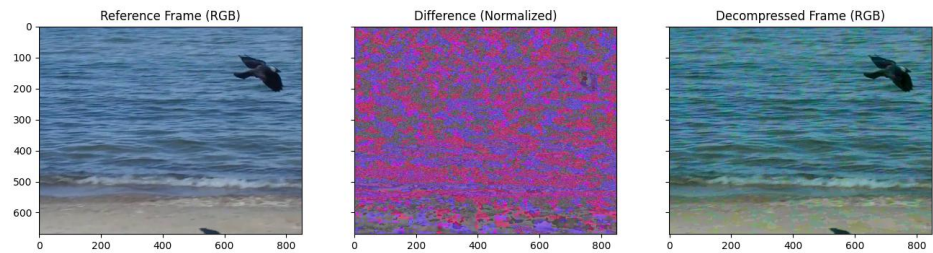
Subsampling: 4:1:1, Divider: 8, Method: None

File:clip_5.mp4, subsampling=4:1:1, divider=8, KeyFrame=4



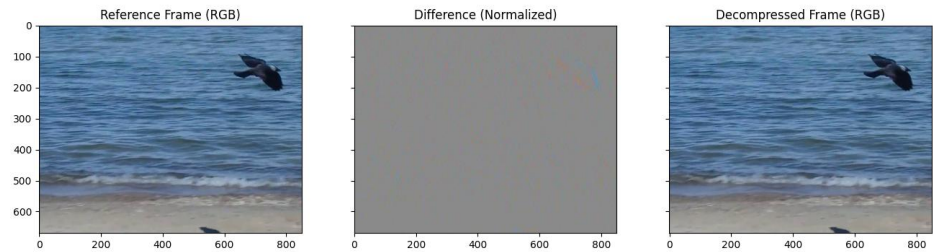
Subsampling: 4:1:1, Divider: 16, Method: None

File:clip_5.mp4, subsampling=4:1:1, divider=16, KeyFrame=4



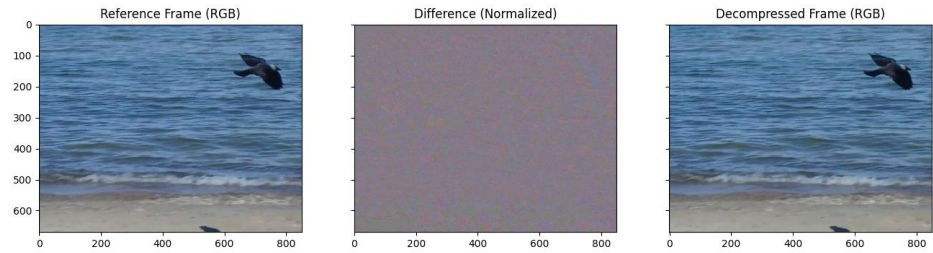
Subsampling: 4:1:0, Divider: 1, Method: None

File:clip_5.mp4, subsampling=4:1:0, divider=1, KeyFrame=4



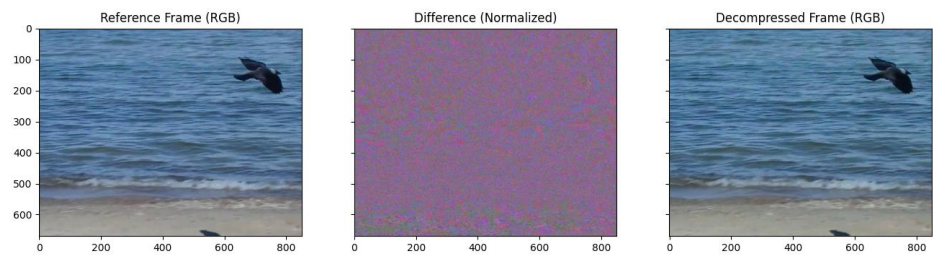
Subsampling: 4:1:0, Divider: 2, Method: None

File:clip_5.mp4, subsampling=4:1:0, divider=2, KeyFrame=4



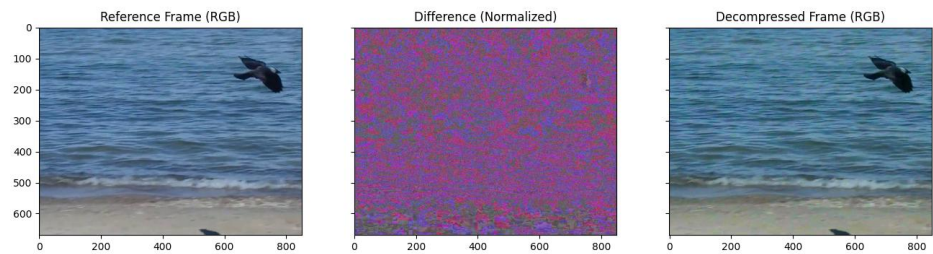
Subsampling: 4:1:0, Divider: 4, Method: None

File:clip_5.mp4, subsampling=4:1:0, divider=4, KeyFrame=4



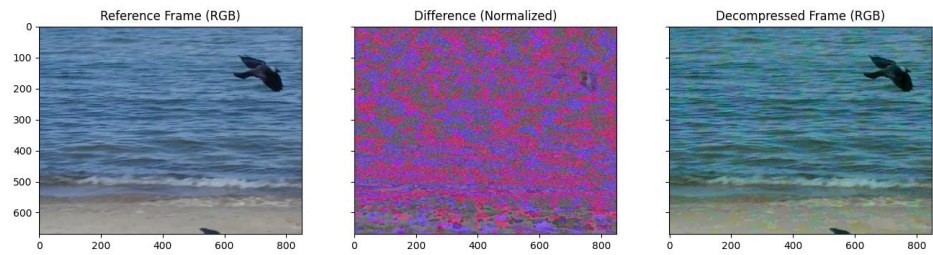
Subsampling: 4:1:0, Divider: 8, Method: None

File:clip_5.mp4, subsampling=4:1:0, divider=8, KeyFrame=4



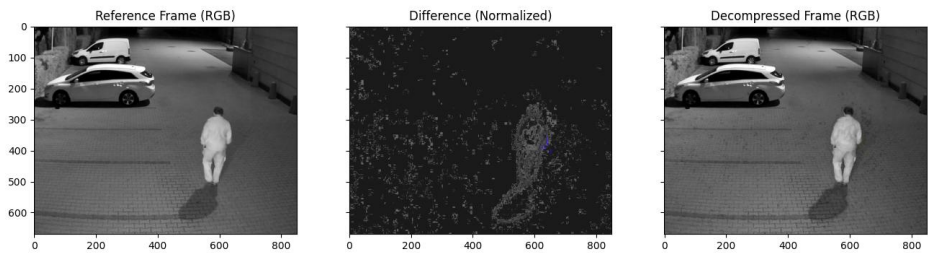
Subsampling: 4:1:0, Divider: 16, Method: None

File:clip_5.mp4, subsampling=4:1:0, divider=16, KeyFrame=4



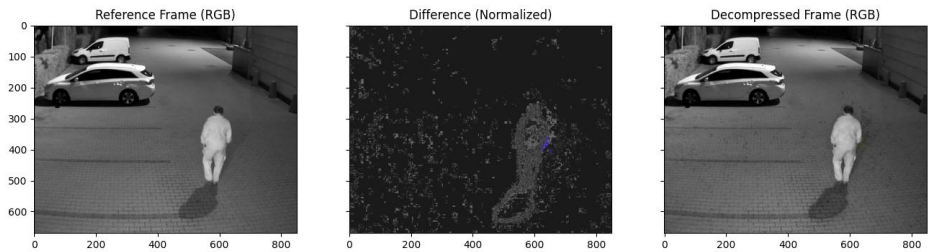
Subsampling: 4:1:0, Divider: 16, Method: ByteRun

File:clip_3.mp4, subsampling=4:1:0, divider=16, KeyFrame=2, Method=ByteRun



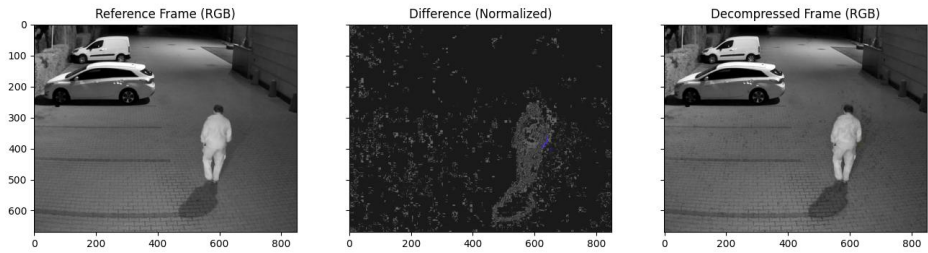
Subsampling: 4:1:0, Divider: 16, Method: ByteRun

File:clip_3.mp4, subsampling=4:1:0, divider=16, KeyFrame=4, Method=ByteRun



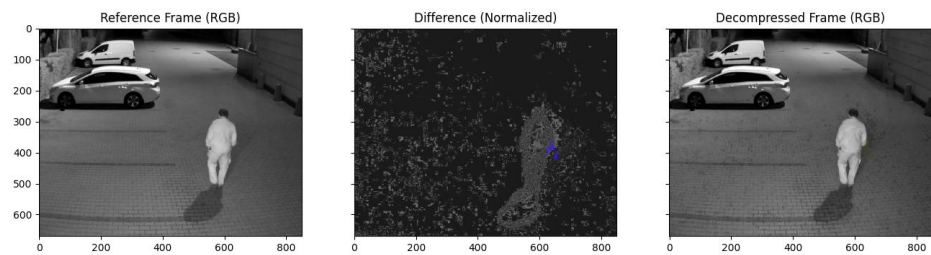
Subsampling: 4:1:0, Divider: 16, Method: ByteRun

File:clip_3.mp4, subsampling=4:1:0, divider=16, KeyFrame=5, Method=ByteRun



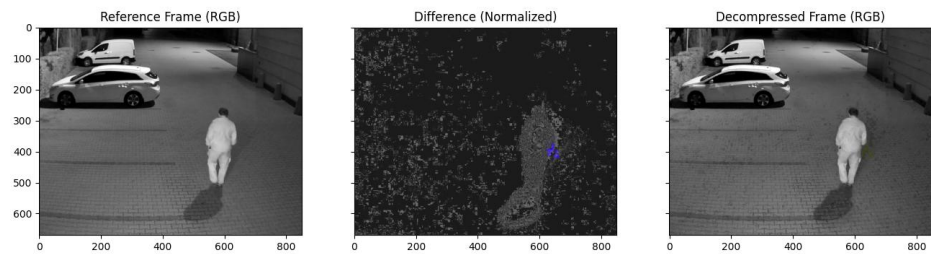
Subsampling: 4:1:0, Divider: 16, Method: ByteRun

File:clip_3.mp4, subsampling=4:1:0, divider=16, KeyFrame=6, Method=ByteRun



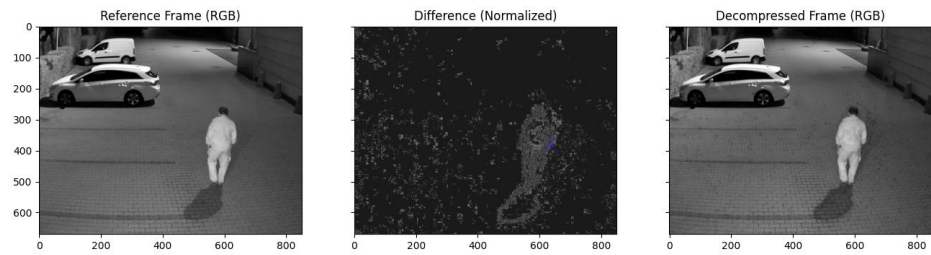
Subsampling: 4:1:0, Divider: 16, Method: ByteRun

File:clip_3.mp4, subsampling=4:1:0, divider=16, KeyFrame=8, Method=ByteRun



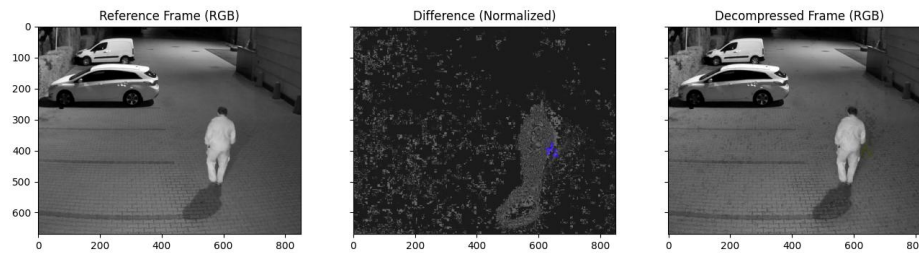
Subsampling: 4:1:0, Divider: 16, Method: ByteRun

File:clip_3.mp4, subsampling=4:1:0, divider=16, KeyFrame=9, Method=ByteRun



Subsampling: 4:1:0, Divider: 16, Method: ByteRun

File: clip_3.mp4, subsampling=4:1:0, divider=16, KeyFrame=10, Method=ByteRun



Dane bazowe dla filmu:

ile = 100

plot_frames = np.array([47])

auto_pause_frames = np.array([25])

subsampling = "4:1:0"

dzielnik = 16

ROI = [[150,800, 150, 800]]

Wnioski:

Zwiększanie odległości początkowo poprawiało stopień kompresji dzięki lepszej redukcji danych w klatkach niekluczowych. Najlepsze wyniki uzyskano przy odległościach 4 i 5 klatek, gdzie uzyskano najlepszy stosunek jakości materiału po dekompresji, a wydajnością kompresji. Przy większych odstępach skuteczność spadała ze względu na narastające różnice między klatkami.

Czy na podstawie waszych wyników opłaca się kompresować Luminancję, poprzez użycie kompresji strumieniowej? Jeżeli tak to dla jakich klatek (kluczowych, niekluczowych)?

Na podstawie otrzymanych wyników stwierdzam iż opłacalne jest kompresowanie Luminancji w przypadku klatek niekluczowych, ponieważ klatki niekluczowe zawierają głównie dane, które często mają długie ciągi powtarzających się wartości przez co idealnie nadają się do kompresji.